

Delusions of Grandeur, and Their Benefits (and Hazards)

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Abstract

We study a population-wide tournament in which agents, who care both about their absolute and relative wealths, experiment by searching over correlated objects. We explore the role of the agents' beliefs about the environment; namely, the stochastic processes corresponding to their experimentation. We find that although optimism leads to higher output, it also produces greater inequality. We connect these observations with empirical evidence suggesting a positive relationship between inequality and entrepreneurship.

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The lottery of the law, therefore, is very far from being a perfectly fair lottery; and that, as well as many other liberal and honourable professions, is, in point of pecuniary gain, evidently under-recompenced.

Those professions keep their level, however, with other occupations, and, notwithstanding these discouragements, all the most generous and liberal spirits are eager to crowd into them. Two different causes contribute to recommend them. First, the desire of the reputation which attends upon superior excellence in any of them; and, secondly, the natural confidence which every man has more or less, not only in his own abilities, but in his own good fortune.

...

The over-weening conceit which the greater part of men have of their own abilities, is an ancient evil remarked by the philosophers and moralists of all ages. Their absurd presumption in their own good fortune, has been less taken notice of. It is, however, if possible, still more universal. There is no man living who, when in tolerable health and spirits, has not some share of it. The chance of gain is by every man more or less over-valued, and the chance of loss is by most men under-valued.

— Smith (1776, Book I, Ch. X)

1 Introduction

Every entrepreneur understands the odds in the third person and rejects them in the first. Every new venture is an argument that its founder is special. The product will find demand, the experiment will work, the market will arrive, or the next iteration will be the one that turns the corner. The social consequences of these delusions of grandeur are not obvious. The same belief that leads an entrepreneur to persist too long may also be the belief that keeps her searching long enough to discover something valuable.

We study the ramifications of entrepreneurial overconfidence in a model of

competition by innovators. Our answer has two parts. Overconfidence raises output because it makes entrepreneurs harder to discourage. They continue after bad news, and so they find better opportunities. But overconfidence also raises inequality. The additional persistence generated by optimistic beliefs does not improve all outcomes equally, raising the upper tail by more than the rest of the distribution. Thus, **delusions of grandeur are productive, but their benefits are unevenly distributed.**

In our model, there is a continuum of *ex-ante* identical entrepreneurs. Each entrepreneur experiments over time and may stop whenever she chooses. If she stops, she implements the best option she has found so far. Continuing is costly, and payoffs depend both on absolute success and on relative success: an entrepreneur values the quality of her own best discovery, but she also values her rank in the population. The distribution of other entrepreneurs' outcomes is, therefore, an equilibrium object and determines how valuable a given discovery is in relative terms.

Crucially, our model also incorporates a simple asymmetry in beliefs. In actuality, no entrepreneur is special: all entrepreneurs face the same true search technology. Subjectively, however, each entrepreneur believes that her own project is special: formally, each entrepreneur believes that her own search process has positive drift, even though the true drift is zero. Moreover, she knows that other entrepreneurs also think they are special and that their decisions reflect this confidence. But she believes that their confidence is misplaced, while her own is warranted. Everyone thinks she is special, everyone knows that everyone else thinks she is special, and everyone thinks that everyone else is wrong.¹

The decision problem each entrepreneur solves (at equilibrium) has a transpar-

¹This premise is consistent with evidence on optimism, overconfidence, and self-assessed ability among entrepreneurs and entrants; see, among others, [Camerer and Lovo \(1999\)](#), [de Meza and Southey \(1996\)](#), [Arabsheibani et al. \(2000\)](#), [Koellinger et al. \(2007\)](#), [Puri and Robinson \(2007\)](#), [Dawson et al. \(2014\)](#), and [Koudstaal et al. \(2016\)](#).

ent form. An entrepreneur does not stop simply because the current state of her project is bad—after all, she can always fall back on the best option she has already discovered. What matters is how far the current project has fallen below that personal best, and so an entrepreneur keeps going until this disappointment becomes too large. Equilibrium behavior is summarized by a tolerance for bad news: how far below her best-so-far outcome an entrepreneur is willing to let the project fall before quitting.

Overconfidence raises this tolerance. A more optimistic entrepreneur thinks the future is more promising, so she is willing to absorb a larger setback before giving up. In equilibrium, this happens throughout the distribution, with more optimistic entrepreneurs searching longer, reaching better personal records, and stopping with better discoveries. Our first main result is, thus, holding fixed the true technology, a more overconfident economy generates a better distribution of realized discoveries (in a first-order stochastic dominance sense). Accordingly, **delusions of grandeur are good:** they raise output and fan the flames of technological growth.

Our second main result is distributional. In particular, the improvement from overconfidence is not a parallel shift. Extra persistence is most valuable for entrepreneurs who end up high in the realized distribution. These entrepreneurs have already survived many earlier setbacks, and the additional willingness to continue helps them push the frontier even farther. Consequently, overconfidence strictly increases the variance of realized outcomes. **Delusions of grandeur are bad:** they lead to greater inequality.

It is important to keep in mind that optimists do not discover more because the true process is better—it is the same in every economy. Optimists discover more because they quit later. Overconfidence acts like a private subsidy to persistence, making discouraging evidence feel less decisive. This raises the average quality of discoveries, but it also increases the time spent experimenting and makes realized

success more unequal.

Our empirical section takes two implications of the model to the data. The first concerns outcomes. If entrepreneurial experimentation raises discoveries while stretching the upper tail, then economies with more entrepreneurship should tend to exhibit more inequality. Consistent with this prediction, we document a positive cross-country relationship between entrepreneurship and inequality, using both the Gini coefficient and the 90/10 percentile ratio. The relationship remains positive after controlling for GDP per capita and population.

The second implication concerns the premise of the model. Our mechanism is not that entrepreneurs simply have a taste for risk. It is that they hold optimistic beliefs about their own prospects. This distinction matters empirically. We find little evidence that country-level measures of risk tolerance, uncertainty avoidance, equity-market participation, lottery sales, fear of failure, or business failure rates explain entrepreneurship. These variables also do not remove the positive relationship between entrepreneurship and inequality. In contrast, survey measures of entrepreneurial self-belief are strongly related to entrepreneurship. In particular, the share of respondents who say that they have the ability to start a business is the strongest predictor in our cross-country exercises. The evidence is only correlational, but it supports the interpretation that entrepreneurship is tied more closely to beliefs about one's own prospects than to (intrinsic) risk appetite.²

We organize the rest of the paper as follows. §1.1 discusses related work. §2 introduces the model. §3 characterizes equilibrium. §4 studies how overconfidence changes output, inequality, and experimentation time. §5 presents the empirical evidence. Proofs of results lie in Appendices A and B.

²This use of survey beliefs connects the paper to the broader survey-based literature on economic perceptions, social position, fairness, and policy views, including Kuziemko et al. (2015), Alesina et al. (2018), Stantcheva (2021), Hvidberg et al. (2023), and Stantcheva (2023).

1.1 Related Work

There is by now a rich literature studying contests in which agents compete by choosing fair gambles,³ or, basically equivalently,⁴ when to stop Brownian motions.⁵ A crucial distinction between these papers and ours is that upon stopping, our agents leave with maximum value their processes hit, whereas in these papers, it is the stopped value of the processes they end with. The Brownian environment corresponds to a search environment with a continuum of correlated items; and in many settings, the “perfect recall” of our environment seems more reasonable than the “no recall” of these papers.

A further connection is to classic all-pay and rank-order contests.⁶ Those models can be understood as contests in which agents choose distributions over output subject to a linear (in distribution) cost. Through that lens, our search problem provides an explicit dynamic foundation for agents’ choices. More broadly, our paper is part of a sizeable literature on incentives for risk taking in tournaments, R&D races, promotion contests, and portfolio-management contests.⁷

The closest technical antecedents of our work are models of search on a correlated path.⁸ In these papers, an agent explores a Brownian or Gaussian environment and later implements the best observation encountered. We share the retrospective-best payoff, but embed the individual stopping problem in a population-

³See, for instance, Fang and Noe (2016, 2022), Wagman and Conitzer (2012), Boleslavsky and Cotton (2015, 2018), Albrecht (2017), Au and Kawai (2020), and Spiegel (2006). In Whitmeyer (2026), the gambles need not be fair.

⁴The Skorokhod embedding theorem is the bridge equating the two.

⁵This problem was first studied by Seel and Strack (2013), with later extensions by Feng and Hobson (2015, 2016a,b), Nutz and Zhang (2022), and Whitmeyer (2023).

⁶Canonical all-pay contest papers include Hillman and Samet (1987), Hillman and Riley (1989), Baye et al. (1996), Barut and Kovenock (1998), Amann and Leininger (1996), and Che and Gale (2000).

⁷See, among others, Dasgupta and Stiglitz (1980), Bhattacharya and Mookherjee (1986), Klette and De Meza (1986), Hvide (2002), Hvide and Kristiansen (2003), Goel and Thakor (2008), Gilpatric (2009), Fang et al. (2020), Basak and Makarov (2015), and Strack (2016).

⁸Refer to, in particular, Urgun and Yariv (2025) and Cetemen et al. (2023).

wide game, making the agents' payoffs endogenous. The paper is also connected to trial-and-error learning and correlated search, where local experimentation reveals information about nearby alternatives.⁹

The paper also speaks to work on optimism, overconfidence, and entrepreneurship. Existing models emphasize entry, financing, managerial selection, and entrepreneurial choice. Our mechanism is different: optimism operates not through who enters, but through how long a fixed population continues experimenting. This distinction is important for welfare. Deluded beliefs can be productive because they sustain search, but hazardous because the gains accrue disproportionately to agents who end up high in the realized distribution.¹⁰

Empirically, the paper is connected to evidence that entrepreneurship is associated with optimistic or overconfident beliefs, and that standard risk-preference measures do not fully account for entrepreneurial behavior. This matches our mechanism: optimism matters because it changes the willingness to continue after bad realizations, not simply because agents have a taste for risk.¹¹ The paper is also related to empirical work on entrepreneurial returns and experimentation. Low average returns, high idiosyncratic risk, and skewed outcomes are consistent with a view of entrepreneurship as experimentation with large upside realizations.¹²

Finally, our empirical discussion relates to the literature on entrepreneurship

⁹See Jovanovic and Rob (1990), Callander (2011b), Callander (2011a), Callander and Matouschek (2019), Callander and Matouschek (2022), and Callander et al. (2025). On correlated attributes, selective learning, and broader correlated-learning environments, see Bardhi (2024) and Bardhi and Callander (2026).

¹⁰On optimism and overconfidence in economic choice, entry, entrepreneurship, and management, see Camerer and Lovo (1999), de Meza and Southey (1996), Bernardo and Welch (2001), Puri and Robinson (2007), Goel and Thakor (2008), and Landier and Thesmar (2009). Related models of motivated or optimal beliefs include Bénabou and Tirole (2002) and Brunnermeier and Parker (2005).

¹¹Evidence on optimism, overconfidence, and entrepreneurship includes Arabsheibani et al. (2000), Koellinger et al. (2007), Puri and Robinson (2007), Dawson et al. (2014), Åstebro (2003), Åstebro et al. (2014), and Koudstaal et al. (2016).

¹²See Hamilton (2000), Moskowitz and Vissing-Jørgensen (2002), Kerr et al. (2014), Manso (2016), and Levine and Rubinstein (2017).

and wealth inequality. Much of that literature studies borrowing constraints, entrepreneurial saving, and business ownership as sources of wealth concentration. We offer a complementary belief-based channel: a society with more entrepreneurial experimentation can generate both higher output and more unequal outcomes, even without ex ante heterogeneity in ability or wealth.¹³

2 Model

There is a continuum of experimenting agents of measure one. Experimentation takes the form of costly search over options: each agent runs an independent driftless Brownian process $X_t = \sigma W_t$, where W is a standard Brownian motion and $\sigma > 0$. Let $M_t := \sup_{0 \leq s \leq t} X_s$ denote the best option discovered by the agent by time t . Agents are misspecified: for perceived drift $\mu > 0$, each agent evaluates stopping times under the perceived law $\mathbb{P}^\mu$, under which $X_t = \mu t + \sigma W_t^\mu$. The parameter μ is the level of *optimism*.

Experimentation is costly: each agent incurs a common flow cost $c > 0$. At any time, an agent may stop experimenting, in which case she receives the best option she has seen so far, M_t . Agents care about both the absolute quality of the best option they find and their rank in the population. Agents evaluate rank through a conjectured rank schedule: a cumulative distribution function K on $[0, \infty)$. Given K , define $\phi_K(m) := m + y(K(m))$, where $y \in C^2([0, 1])$, $y' > 0$, $y'' \leq 0$, and y' is bounded. An agent chooses a stopping time τ to solve

$$\sup_{\tau} \mathbb{E}^\mu [\phi_K(M_\tau) - c\tau].$$

At equilibrium, the conjectured schedule K equals the realized distribution of stopped maxima under the true law.

¹³On entrepreneurship, wealth, and inequality, see [Evans and Jovanovic \(1989\)](#), [Quadrini \(2000\)](#), [Hurst and Lusardi \(2004\)](#), and [Cagetti and De Nardi \(2006\)](#).

For the verification arguments below, it is useful to describe the problem from an arbitrary current position. If the agent's current best option is m , her current option can be described as $m - z$, where z is the current *drawdown*. Thus, as a random variable, the drawdown is $D_t := M_t - X_t$. Starting from (m, z) means $(M_0, D_0) = (m, z)$, and we write $\mathbb{E}_{m,z}^\mu$ for expectation under the perceived law from this starting point. We say a stopping time τ is *K-admissible* from (m, z) if $\mathbb{E}_{m,z}^\mu[\tau] < \infty$ and $\mathbb{E}_{m,z}^\mu[\phi_K(M_\tau)^+] < \infty$. In our model, agents begin from $(0, 0)$. In the fixed- K stopping problem, we take optimality over K -admissible stopping times.

Next, define $\bar{d}_\mu := \sigma^2 \log(c/(c - \mu))/(2\mu)$, and let

$$\mathcal{M} := \left\{ \mu \in (0, c) : \frac{\mu}{c} \left(1 + \frac{y'(0)}{\bar{d}_\mu} \right) < 1 \right\}.$$

We make the standing assumption $\mu \in \mathcal{M}$, which is a finite-stopping condition. It rules out cases in which the perceived drift and the marginal rank reward are so large that the agent wants to continue forever.

A *drawdown boundary* is a positive function $d : [0, \infty) \rightarrow (0, \infty)$. Given d , define

$$\tau_d := \inf \{ t \geq 0 : M_t - X_t \geq d(M_t) \},$$

and write

$$C_d := \{(m, z) : m \geq 0, 0 < z < d(m)\}.$$

For a conjecture K with $\phi_K \in C^1$, write

$$V_K^\mu(m, z) := \sup_{\tau} \mathbb{E}_{m,z}^\mu[\phi_K(M_\tau) - c\tau],$$

where the supremum is over stopping times that are K -admissible from (m, z) . A function V is *regular* for d with terminal payoff ψ at perceived drift μ if $V \in C^{1,2}(C_d)$, extends continuously with the relevant one-sided derivatives to the bound-

aries, satisfies

$$\frac{\sigma^2}{2} V_{zz}(m, z) - \mu V_z(m, z) - c = 0, \quad \forall (m, z) \in C_d,$$

and obeys *value matching* $V(m, d(m)) = \psi(m)$, *smooth fit* $V_z(m, d(m)^-) = 0$, and the *reflected-boundary condition* $V_m(m, 0^+) + V_z(m, 0^+) = 0$.

Given K , the boundary d is a *regular optimal drawdown boundary* for the fixed- K problem at perceived drift μ if τ_d attains $V_K^\mu(m, z)$ for every (m, z) with $0 \leq z < d(m)$, among stopping times that are K -admissible from (m, z) , and V_K^μ is regular for d with terminal payoff ϕ_K at perceived drift μ .

Definition 2.1. A *regular drawdown equilibrium* at perceived drift $\mu \in \mathcal{M}$ is a pair (K, d) with the following properties. The conjecture K is a cumulative distribution function on $[0, \infty)$ such that $\phi_K \in C^1$. The boundary d is C^1 , positive, bounded, bounded away from zero, and has a finite limit. Given K , the boundary d is a regular optimal drawdown boundary for the fixed- K problem and K is consistent with the agents' equilibrium stopping rule under the true law: $K(m) = P^0(M_{\tau_d} \leq m)$.

3 Equilibrium Characterization

The point of this section is to turn the equilibrium fixed point into a one-dimensional object. The difficulty is that the two equilibrium objects discipline each other. A drawdown boundary d determines, under the true law $\mathbb{P}^0$, how far agents climb before stopping and, hence, what the realized rank distribution is. At the same time, the conjectured rank schedule K determines, under the perceived law $\mathbb{P}^\mu$, how valuable it is for an individual agent to continue searching. The characterization below separates these two roles before putting them back together.

We begin with the distributional side. Suppose agents stop when the current gap between the best option found and the current option reaches $d(M_t)$. Con-

ditional on having reached a record m , a larger value of $d(m)$ makes it less likely that the agent stops before reaching still higher records. In fact, over a small interval $[m, m + \Delta]$, the stopping probability is approximately $\Delta/d(m)$. The survival probability of the stopped maximum is the accumulation of these local exit rates.

Toward stating our first result, given d , define $A_d(m) := \int_0^m \frac{1}{d(r)} dr$, and denote $T_m := \inf\{t \geq 0: M_t \geq m\}$.

Lemma 3.1. *Let drawdown $d \in C^1([0, \infty))$ be positive and locally bounded away from zero. Then $\mathbb{P}^0(T_m < \tau_d) = \exp(-A_d(m))$. If $A_d(m) \rightarrow \infty$ as $m \rightarrow \infty$, then $\tau_d < \infty$ $\mathbb{P}^0$ -almost surely and $\mathbb{P}^0(M_{\tau_d} \geq m) = \exp(-A_d(m))$. Moreover, $K_d(m) := 1 - \exp(-A_d(m))$ is the induced CDF of M_{τ_d} .*

The proof is in [Appendix A.1](#). The lemma says that the inverse drawdown $1/d(m)$ is the local rate at which agents exit before pushing their maximum past m . Thus, a drawdown boundary is not only an individual stopping rule; under the true law, it is also a recipe for the cross-sectional rank distribution.

We next turn to the individual optimality side. Fix a conjectured rank schedule K . If an agent stops with maximum m , her effective prize is $\phi_K(m) = m + y(K(m))$. In [Appendix A.2](#), we show that, within the regular drawdown class, optimality for this fixed- K problem is characterized by the free-boundary equation

$$d'(m) = 1 - \frac{\mu}{c} \frac{\phi'_K(m)}{1 - \exp\left(-\frac{2\mu}{\sigma^2} d(m)\right)}.$$

When K is differentiable, $\phi'_K(m) = 1 + y'(K(m))K'(m)$. This object decomposes in a straightforward manner: the term 1 is the direct marginal value of raising the best option found. The second term $y'(K(m))K'(m)$ is the marginal rank value: a slightly higher maximum improves the agent's position in the population, and this is valuable because $y' > 0$. This is the channel through which relative concerns affect the stopping boundary.

The limiting boundary has a simple interpretation. When the marginal rank

term vanishes, the agent behaves like a single-agent experimenter whose marginal prize from a higher maximum is just one. A finite limiting drawdown must then converge to $\bar{d}_\mu$, where

$$1 - \exp\left(-\frac{2\mu}{\sigma^2}\bar{d}_\mu\right) = \frac{\mu}{c}.$$

This value will be the terminal condition for the equilibrium characterization.

We now impose the consistency necessitated by equilibrium. That is to say, in equilibrium, the conjectured rank schedule must equal the true distribution generated by the optimal boundary: $K = K_d$. From [Lemma 3.1](#), this means $K(m) = 1 - \exp(-A_d(m))$. Rather than solve directly for both K and d , we use $t = A_d(m)$ as the coordinate. In this coordinate, the agent with maximum $m(t)$ has rank $K(m(t)) = 1 - e^{-t}$. Define the quantile-indexed drawdown $\delta(t) := d(m(t))$. Since $A_d(m(t)) = t$, we differentiate to obtain $m'(t)/d(m(t)) = 1$, or simply $m'(t) = \delta(t)$. Consequently, the same function that records the drawdown tolerated at each rank also specifies the slope of the equilibrium quantile function.

The rank term also simplifies after this change of variables. Since $K(m(t)) = 1 - e^{-t}$, we can again differentiate to get $K'(m(t))m'(t) = e^{-t}$. Using $m'(t) = \delta(t)$, we obtain $K'(m(t)) = e^{-t}/\delta(t)$. Therefore, the marginal prize at the maximum $m(t)$ is

$$\phi'_K(m(t)) = 1 + \frac{b(t)}{\delta(t)}, \quad b(t) := e^{-t}y'(1 - e^{-t}).$$

Substituting this expression into the fixed- K boundary equation at $m = m(t)$ and using $\delta'(t) = d'(m(t))m'(t) = d'(m(t))\delta(t)$, any regular equilibrium must satisfy

$$\delta'(t) = \delta(t) - \frac{\mu}{c} \frac{\delta(t) + b(t)}{1 - \exp\left(-\frac{2\mu}{\sigma^2}\delta(t)\right)}.$$

This is the key reduction. The equilibrium fixed point in the two objects K and d becomes a terminal-value problem for the single function δ . Our theorem shows that this reduction is complete: the scalar equation has a unique bounded posi-

tive solution, and that solution reconstructs the unique regular drawdown equilibrium.

Theorem 3.2. *Fix $\mu \in \mathcal{M}$. There exists a unique regular drawdown equilibrium at perceived drift μ . It is characterized by the unique bounded positive C^1 solution δ_μ of*

$$\delta'_\mu(t) = \delta_\mu(t) - \frac{\mu}{c} \frac{\delta_\mu(t) + b(t)}{1 - \exp\left(-\frac{2\mu}{\sigma^2} \delta_\mu(t)\right)}$$

satisfying $\lim_{t \rightarrow \infty} \delta_\mu(t) = \bar{d}_\mu$, where $b(t) := e^{-t} y'(1 - e^{-t})$. Moreover, if $\alpha_\mu := 1 + y'(0)/\bar{d}_\mu$ and $d_\mu^+ := \sigma^2 \log(c/(c - \mu\alpha_\mu))/(2\mu)$, then $\bar{d}_\mu \leq \delta_\mu(t) \leq d_\mu^+$ for all $t \geq 0$. The equilibrium quantile map $m_\mu(t) := K_\mu^{-1}(1 - e^{-t})$ satisfies $m'_\mu(t) = \delta_\mu(t)$.

Equivalently, once δ_μ is known, we can recover the equilibrium by setting $m_\mu(t) = \int_0^t \delta_\mu(s) ds$, $K_\mu(m_\mu(t)) = 1 - e^{-t}$, and $d_\mu(m_\mu(t)) = \delta_\mu(t)$.

The proof is in [Appendix A.3](#). The theorem supplies a compact way of understanding the equilibrium. At rank $1 - e^{-t}$, the agent tolerates a drawdown of size $\delta_\mu(t)$. Because $m'_\mu(t) = \delta_\mu(t)$, the same object also determines how spread out equilibrium outcomes are across adjacent ranks. The function $b(t)$ is the marginal rank incentive in this coordinate. The factor $y'(1 - e^{-t})$ is the marginal utility of rank, while the factor e^{-t} converts a change in the hazard coordinate into a change in rank. Since $b(t) \rightarrow 0$, rank incentives vanish at the top of the distribution, and the equilibrium drawdown converges to the single-agent benchmark $\bar{d}_\mu$.

4 Comparative Statics

This section contains the main comparative statics. We compare economies that differ only in perceived drift μ . Crucially we evaluate the underlying process and all realized outcomes under the true law $\mathbb{P}^0$. The characterization in [Section 3](#) makes the comparison transparent. Each equilibrium is summarized by the quantile-indexed drawdown δ_μ , and the equilibrium quantile map satisfies

$m'_\mu(t) = \delta_\mu(t)$. Thus, once we know how optimism changes δ_μ , we immediately know how it changes the distribution of stopped maxima.

The main force is simple. A more optimistic agent believes continued experimentation is more promising, so she is willing to continue after a larger decline from her personal best. In equilibrium, this greater tolerance for setbacks appears at every finite hazard-time quantile.

Lemma 4.1. *Let $\mu_1, \mu_2 \in \mathcal{M}$ with $\mu_1 < \mu_2$. Then $\delta_{\mu_2}(t) > \delta_{\mu_1}(t)$ for every finite $t \geq 0$. Equivalently, $m'_{\mu_2}(t) > m'_{\mu_1}(t)$ for every finite $t \geq 0$.*

The proof is in the appendix. The intuition comes directly from the equation in [Theorem 3.2](#). The rank term $b(t) = e^{-t}y'(1 - e^{-t})$ is the same across economies, while a larger μ raises the perceived value of continued search and also raises the limiting single-agent drawdown $\bar{d}_\mu$. Our argument in the appendix shows that this ordering cannot be undone at any finite t . Since $m'_\mu(t) = \delta_\mu(t)$, the more optimistic economy has a steeper true-population quantile map everywhere.

Now let $M_\mu := M_{\tau_{d_\mu}}$ denote the equilibrium stopped maximum under the true law. The previous lemma compares quantile slopes. The next lemma converts that comparison into a distributional statement by representing stopped maxima directly in quantile coordinates.

Lemma 4.2. *Let $Z \sim \text{Exp}(1)$. Then under $\mathbb{P}^0$, $M_\mu \stackrel{d}{=} m_\mu(Z)$.*

This follows immediately from $K_\mu(m_\mu(t)) = 1 - e^{-t}$. The random variable Z is the hazard-time rank of a randomly chosen agent in the true population, and $m_\mu(Z)$ converts that rank into the realized stopped maximum. This representation lets us compare different economies using the same random rank draw Z .

Theorem 4.3. *Let $\mu_1, \mu_2 \in \mathcal{M}$ with $\mu_1 < \mu_2$. In the unique regular drawdown equilibria,*

$$M_{\mu_2} \succeq_{\text{FOSD}} M_{\mu_1}, \quad \text{and} \quad \text{Var}^0(M_{\mu_2}) > \text{Var}^0(M_{\mu_1}).$$

The proof is in the appendix. The idea is short. By [Lemma 4.1](#), m_{μ_2} is everywhere steeper than m_{μ_1} . Since both quantile maps start at zero, $m_{\mu_2}(t) > m_{\mu_1}(t)$ for every $t > 0$. Using the representation in [Lemma 4.2](#), the more optimistic economy, therefore, has a higher stopped maximum at every positive quantile, which drives first-order stochastic dominance and a higher mean.

The variance result comes from the same geometry. Let $\Delta(t) := m_{\mu_2}(t) - m_{\mu_1}(t)$. Since $m'_{\mu_2}(t) > m'_{\mu_1}(t)$, the gain $\Delta(t)$ is increasing in t . Under the common- Z representation, the more optimistic economy can be written as $m_{\mu_1}(Z) + \Delta(Z)$. Thus, the optimism-induced gain is largest precisely for agents who are already high in the distribution. Optimism raises the whole distribution, but it raises the upper quantiles by more in absolute terms, so dispersion increases.

The stopped maximum M_μ is the output object compared above, but the same equilibrium boundary also determines how long agents experiment in true time. Define

$$\tau_\mu := \tau_{d_\mu} = \inf\{t \geq 0: M_t - X_t \geq d_\mu(M_t)\}.$$

In the appendix, we derive the useful identity

$$\mathbb{E}^0[\tau_\mu] = \frac{1}{\sigma^2} \mathbb{E}^0[d_\mu(M_\mu)^2] = \frac{1}{\sigma^2} \int_0^\infty \delta_\mu(t)^2 e^{-t} dt.$$

This formula clarifies the source of the output effect. Optimists do not obtain higher stopped maxima because the true process has a higher drift; all realized comparisons are under $\mathbb{P}^0$. They obtain higher stopped maxima because they tolerate larger drawdowns and so search longer. The weight e^{-t} is the density of the hazard-time rank Z , while the term $\delta_\mu(t)^2/\sigma^2$ reflects the quadratic time scale of Brownian fluctuations.

Proposition 4.4. *If $\mu_1, \mu_2 \in \mathcal{M}$ and $\mu_1 < \mu_2$, then $\mathbb{E}^0[\tau_{\mu_2}] > \mathbb{E}^0[\tau_{\mu_1}]$.*

The result follows from the identity above and [Lemma 4.1](#): higher optimism raises $\delta_\mu(t)$ at every finite t , so it raises the population average of squared draw-

down tolerances. Thus, the same behavioral change that raises stopped maxima also lengthens true experimentation time.

Together, these results capture both the benefit and the hazard of optimism. Greater optimism raises realized output under the true law, but it does so by lengthening search and steepening the equilibrium quantile function. The gains are, therefore, uneven across ranks: the whole distribution improves, while dispersion increases.

5 Empirical Evidence

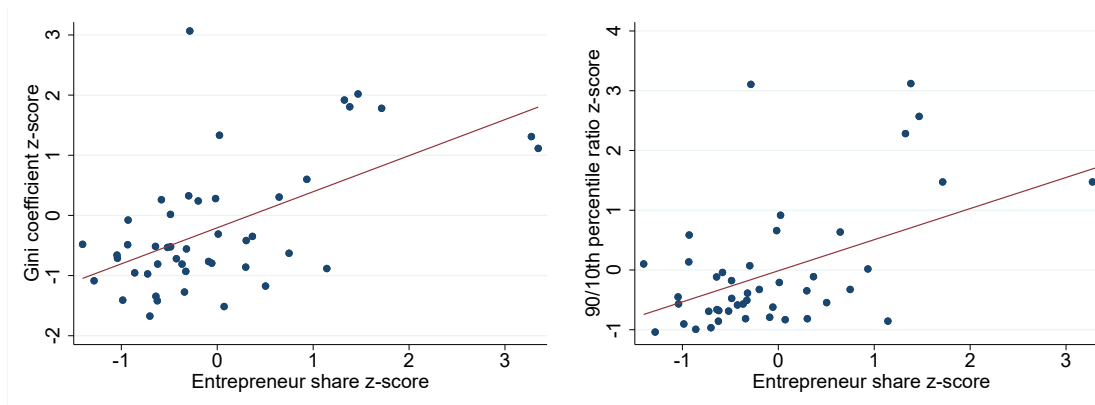


Figure 1: Entrepreneurship and Inequality

These figures show country-level correlations between standardized measures of inequality as measured by the Gini index (left panel) or 90/10th percentile ratio (right panel).

	Gini coefficient				90/10th percentile ratio			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Entrepreneurship share	0.599*** (0.130)	0.454** (0.181)	0.620*** (0.123)	0.469** (0.178)	0.521*** (0.133)	0.379* (0.200)	0.531*** (0.132)	0.386* (0.202)
Log GDP per capita		-0.409** (0.185)		-0.334* (0.189)		-0.223 (0.205)		-0.187 (0.215)
Log population			0.186** (0.074)	0.123 (0.085)			0.099 (0.079)	0.059 (0.096)
Observations	46	34	46	34	46	34	46	34
R ²	0.327	0.350	0.415	0.392	0.259	0.183	0.285	0.193

Table 1: Entrepreneurship and Inequality

This table shows a positive relationship between inequality and entrepreneurship after controlling for GDP per capita and population.

	(1)	(2)	(3)	(4)
Willingness to take risks	0.064 (0.172)			
Uncertainty avoidance index		0.168 (0.169)		
Equity share of household assets			0.292 (0.233)	
Lottery sales per capita				-0.370* (0.197)
Observations	33	46	26	34
R ²	0.004	0.022	0.061	0.099

Table 2: Entrepreneurship and Risk Aversion

This table shows that there is no statistically or economically significant relationship between country-level measures of entrepreneurship and risk taking.

	(1)	(2)	(3)	(4)
Entrepreneurship share	0.593*** (0.165)	0.598*** (0.134)	0.505** (0.188)	0.623*** (0.159)
Willingness to take risks	0.289 (0.173)			
Uncertainty avoidance index		0.078 (0.162)		
Lottery sales per capita			-0.309 (0.220)	
Equity share of household assets				0.171 (0.187)
Observations	32	44	34	26
R ²	0.361	0.344	0.293	0.454

Table 3: Entrepreneurship, Inequality, and Risk Aversion

This table shows that the relationship between inequality and entrepreneurship holds even after controlling for measures of risk preferences.

	(1)	(2)	(3)	(4)	(5)
Belief in own abilities	0.561*** (0.120)				0.538*** (0.198)
Others think you are innovative		0.406*** (0.132)			-0.033 (0.221)
Think entrepreneur is a good career choice			0.320** (0.140)		0.025 (0.153)
Know someone who recently started a business				0.373*** (0.134)	0.085 (0.157)
Observations	50	50	49	50	49
R ²	0.314	0.165	0.100	0.139	0.327

Table 4: Drivers of Entrepreneurship

This table shows that the relationship between entrepreneurship and self-reported measures of beliefs and perceptions related to entrepreneurship.

	(1)	(2)	(3)	(4)	(5)
Fear of failure	0.025 (0.144)				0.005 (0.147)
Motivation for great wealth		-0.126 (0.143)			-0.252 (0.163)
Assign high social status to entrepreneurs			0.112 (0.146)		0.035 (0.153)
Business failure rate				0.185 (0.142)	0.280 (0.168)
Observations	50	50	49	50	49
R ²	0.001	0.016	0.012	0.034	0.088

Table 5: Preferences and Entrepreneurship

This table shows that entrepreneurship has no meaningful relationship with several alternative measures of risk preferences, including actual business failure rates.

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A Equilibrium Characterization (Section 3) Proofs

A.1 Proof of Lemma 3.1

Proof of Lemma 3.1. Define $G(x) := \exp\left(\int_0^x \frac{1}{d(r)} dr\right)$ and $g(x) := G'(x) = G(x)/d(x)$. Recall that $D_t := M_t - X_t$, and consider

$$N_t := G(M_t) - g(M_t)D_t.$$

Claim A.1. N is a local martingale under $\mathbb{P}^0$.

Proof. The running maximum M_t can increase only when $X_t = M_t$, i.e., only when $D_t = 0$. Equivalently, the Stieltjes measure dM_t assigns mass only to times at which $D_t = 0$, and so $D_t dM_t = 0$. Since $dD_t = dM_t - dX_t$, we have

$$dN_t = G'(M_t)dM_t - g(M_t)dD_t - D_t g'(M_t)dM_t = g(M_t)dX_t,$$

so N is a local martingale under $\mathbb{P}^0$. ■

Next, we establish the first conclusion in the lemma.

Claim A.2. For every $m \geq 0$, $\mathbb{P}^0(T_m < \tau_d) = \exp(-A_d(m))$.

Proof. Stop N at $T_m \wedge \tau_d$. On this stopped interval, $M_t \leq m$ and $0 \leq D_t \leq d(M_t)$, so $0 \leq N_{t \wedge T_m \wedge \tau_d} \leq G(m)$. Accordingly, the stopped local martingale is bounded, hence, a true martingale; and optional stopping yields $\mathbb{E}^0[N_{t \wedge T_m \wedge \tau_d}] = N_0$. Since $T_m < \infty$ $\mathbb{P}^0$ -almost surely, letting $t \rightarrow \infty$, we obtain $\mathbb{E}^0[N_{T_m \wedge \tau_d}] = N_0$. On $\{T_m < \tau_d\}$,

the continuous process X reaches the new record level m exactly at time T_m . Hence, $M_{T_m} = X_{T_m} = m$, so $D_{T_m} = 0$ and $N_{T_m} = G(m)$. On $\{\tau_d < T_m\}$, $D_{\tau_d} = d(M_{\tau_d})$, so $N_{\tau_d} = 0$. Since $d > 0$, the two stopping times cannot coincide. Accordingly,

$$1 = N_0 = \mathbb{E}^0 \left[N_{T_m \wedge \tau_d} \right] = G(m) \mathbb{P}^0 (T_m < \tau_d).$$

Thus, $\mathbb{P}^0(T_m < \tau_d) = G(m)^{-1} = \exp(-A_d(m))$. ■

We finish with the last two conclusions of the result.

Claim A.3. *If $A_d(m) \rightarrow \infty$, then $\tau_d < \infty$ $\mathbb{P}^0$ -almost surely.*

Proof. Under $\mathbb{P}^0$, $M_t \rightarrow \infty$ almost surely. Therefore, $\{\tau_d = \infty\} = \bigcap_{n \geq 1} \{T_n < \tau_d\}$, and [Claim A.2](#) delivers

$$\mathbb{P}^0(\tau_d = \infty) = \lim_{n \rightarrow \infty} \mathbb{P}^0(T_n < \tau_d) = \lim_{n \rightarrow \infty} \exp(-A_d(n)) = 0. \quad \blacksquare$$

Claim A.4. *The induced distribution of M_{τ_d} has CDF $K_d(m) := 1 - \exp(-A_d(m))$.*

Proof. Since $\tau_d < \infty$ almost surely, $\{T_m < \tau_d\} = \{M_{\tau_d} \geq m\}$. Indeed, $T_m < \tau_d$ implies $M_{\tau_d} \geq m$, while $M_{\tau_d} \geq m$ implies $T_m \leq \tau_d$, and equality is impossible by the no-coincidence argument above. Therefore, $\mathbb{P}^0(M_{\tau_d} \geq m) = \exp(-A_d(m))$. Since A_d is continuous, this survival function is continuous, so M_{τ_d} has no atoms. Accordingly, $\mathbb{P}^0(M_{\tau_d} \leq m) = 1 - \exp(-A_d(m))$. ■

This proves the lemma. ■

A.2 Two Auxiliary Results

[Lemma 3.1](#) maps a drawdown boundary into the true-law distribution of stopped maxima. We now go the other direction. First, in [Proposition A.5](#), for a fixed conjectured rank schedule K , we characterize a boundary that is optimal for the

agent under the perceived law. Then, in [Lemma A.11](#) we show conversely that any (regular) optimal boundary indeed satisfies this characterization.

Recall that the effective prize is $\phi_K(m) := m + y(K(m))$. The verification theorem is stated for a generic prize function ϕ . Write $D_t := M_t - X_t$. By Skorokhod reflection [Karatzas and Shreve \(1991, Ch. 3, Sec. 6.C\)](#),

$$D_t = D_0 - \mu t - \sigma W_t^\mu + L_t \quad \text{and} \quad M_t = M_0 + L_t, \quad (1)$$

where L_t increases only when $D_t = 0$.

Proposition A.5. *Fix $\mu \in (0, c)$, and let $\phi \in C^1([0, \infty))$ be increasing and bounded below, with $\sup_m \phi'(m) < c/\mu$. Suppose an agent's drawdown boundary d is C^1 , positive, bounded, bounded away from zero, and satisfies $d(m) \rightarrow d_\infty < \infty$. If d solves*

$$d'(m) = 1 - \frac{\mu}{c} \frac{\phi'(m)}{1 - \exp\left(-\frac{2\mu}{\sigma^2} d(m)\right)}, \quad (2)$$

then τ_d is optimal from every (m, z) with $0 \leq z < d(m)$, among stopping times τ satisfying $\mathbb{E}_{m,z}^\mu[\tau] < \infty$ and $\mathbb{E}_{m,z}^\mu[\phi(M_\tau)^+] < \infty$. Moreover, the associated fixed- ϕ value function is regular for d with terminal payoff ϕ at perceived drift μ . If, in addition, $\phi'(m) \rightarrow 1$, then $d_\infty = \bar{d}_\mu$.

Proof of [Proposition A.5](#). Let $a := 2\mu/\sigma^2$. For $0 \leq z \leq d(m)$, define

$$V(m, z) := \phi(m) + \frac{c}{\mu} \left[d(m) - z - \frac{1 - \exp(-a(d(m) - z))}{a} \right],$$

and for $z \geq d(m)$, set $V(m, z) := \phi(m)$. Again, we proceed by going through a sequence of claims. Throughout this proof, admissible means satisfying the two integrability conditions in the statement.

Claim A.6. *The function V is $C^{1,2}(C_d)$, extends continuously with the relevant one-sided derivatives to the boundaries, and satisfies $V \geq \phi$, the HJB equation in the contin-*

uation region, value matching, smooth fit, and the reflected-boundary condition.

Proof. Inside C_d , we differentiate to obtain

$$V_m(m, z) = \phi'(m) + \frac{c}{\mu} d'(m) [1 - \exp(-a(d(m) - z))],$$

$$V_z(m, z) = \frac{c}{\mu} [\exp(-a(d(m) - z)) - 1], \quad \text{and} \quad V_{zz}(m, z) = \frac{2c}{\sigma^2} \exp(-a(d(m) - z)).$$

Since $\phi, d \in C^1$, these derivatives are continuous on C_d , and the relevant one-sided boundary derivatives exist continuously. Since $x - (1 - \exp(-ax))/a \geq 0$ for $x \geq 0$, $V \geq \phi$. At $z = d(m)$, $V(m, d(m)) = \phi(m)$ and $V_z(m, d(m)^-) = 0$. Using the expressions for V_z and V_{zz} , we also have $(\sigma^2/2)V_{zz} - \mu V_z - c = 0$ on C_d . At the reflecting boundary $z = 0$,

$$V_m(m, 0^+) + V_z(m, 0^+) = \phi'(m) + \frac{c}{\mu} [1 - \exp(-ad(m))] [d'(m) - 1].$$

Thus, the boundary ODE (2) is exactly the condition $V_m(m, 0^+) + V_z(m, 0^+) = 0$. ■

Claim A.7. *The process $V(M_t, D_t) - ct$ is a local supermartingale under $\mathbb{P}^\mu$.*

Proof. Using the reflected dynamics stated in (1), the local-time term at $z = 0$ is proportional to $V_m + V_z$. The generalized Itô formula with local time on surfaces (Peskir, 2007, Theorem 2.1) applies to the pasted value function. The HJB kills the drift in the continuation region, smooth fit kills the boundary local time at $z = d(m)$, and the condition $V_m(m, 0) + V_z(m, 0) = 0$ kills the reflection term at $z = 0$. In the stopping region $z > d(m)$, the pasted value is $V(m, z) = \phi(m)$, so $V_z = V_{zz} = 0$, M_t is locally constant away from $z = 0$, and the drift of $V(M_t, D_t) - ct$ is $-c$. Hence, $V(M_t, D_t) - ct$ is a local supermartingale. ■

Claim A.8. *For every admissible stopping time τ ,*

$$\mathbb{E}_{m,z}^\mu [V(M_\tau, D_\tau) - c\tau] \leq V(m, z).$$

Proof. Fix an arbitrary starting state (m, z) with $0 \leq z < d(m)$. Let

$$\eta_N := \inf\{t \geq 0: M_t + D_t \geq N\} \quad \text{and} \quad \tau_N := \tau \wedge N \wedge \eta_N.$$

For $N > m + z$, the stopping time τ_N is bounded by N , and before η_N we have $M_t + D_t \leq N$. Thus, $V(M_{t \wedge \tau_N}, D_{t \wedge \tau_N}) - c(t \wedge \tau_N)$ is bounded. Consequently, from optional sampling

$$\mathbb{E}_{m,z}^\mu [V(M_{\tau_N}, D_{\tau_N}) - c\tau_N] \leq V(m, z).$$

Because d is bounded, $V \leq \phi + c \sup_m d(m)/\mu$, while $V \geq \phi$ and ϕ is bounded below. Because $\tau_N \uparrow \tau$ and the sample paths are continuous, $M_{\tau_N} \rightarrow M_\tau$ and $D_{\tau_N} \rightarrow D_\tau$.

Let $C := c \sup_m d(m)/\mu$. Since $V \leq \phi + C$, $M_{\tau_N} \leq M_\tau$, and ϕ is increasing, the positive parts are bounded by the integrable random variable $\phi(M_\tau)^+ + C$. Since $V \geq \phi$ and ϕ is bounded below, $V(M_{\tau_N}, D_{\tau_N}) - c\tau_N \geq \inf_m \phi(m) - c\tau$, whose right-hand side is integrable. Fatou's lemma, therefore, yields

$$\mathbb{E}_{m,z}^\mu [V(M_\tau, D_\tau) - c\tau] \leq \liminf_{N \rightarrow \infty} \mathbb{E}_{m,z}^\mu [V(M_{\tau_N}, D_{\tau_N}) - c\tau_N] \leq V(m, z). \quad \blacksquare$$

Since $V \geq \phi$, [Claim A.8](#) implies

$$\mathbb{E}_{m,z}^\mu [\phi(M_\tau) - c\tau] \leq V(m, z)$$

for every admissible stopping time τ . It remains to show that τ_d is admissible and attains this upper bound.

Claim A.9. *The stopping time τ_d is admissible.*

Proof. Let $D := \sup_m d(m)$, and define $\tau_D := \inf\{t \geq 0: D_t \geq D\}$. Since $d \leq D$, we have $\tau_d \leq \tau_D$. By Lehoczky's fixed-drawdown formula ([Lehoczky, 1977](#), Eq. (5)),

when the process starts with no drawdown, $D_0 = 0$,

$$\mathbb{P}^\mu(M_{\tau_D} - M_0 \geq r) = \exp(-\rho_D r), \quad \text{where} \quad \rho_D := \frac{2\mu}{\sigma^2[\exp(2\mu D/\sigma^2) - 1]}.$$

Starting from any drawdown $z \in [0, D]$ can only reduce the chance of reaching new maxima before hitting D , so the same exponential tail bound applies.

Moreover, the killed reflected drawdown process on $[0, D]$ has finite mean. Indeed, if $h(z) := \mathbb{E}_z^\mu[\tau_D]$, then

$$\frac{\sigma^2}{2}h''(z) - \mu h'(z) = -1, \quad \text{with} \quad h'(0) = 0, \quad \text{and} \quad h(D) = 0.$$

Solving this,

$$\mathbb{E}_z^\mu[\tau_D] = \frac{\sigma^2}{2\mu^2} \left[\exp\left(\frac{2\mu D}{\sigma^2}\right) - \exp\left(\frac{2\mu z}{\sigma^2}\right) \right] - \frac{D-z}{\mu} < \infty.$$

Thus, $\mathbb{E}_{m,z}^\mu[\tau_D] < \infty$, and since $\tau_d \leq \tau_D$, $\mathbb{E}_{m,z}^\mu[\tau_d] < \infty$. Since $M_{\tau_d} \leq M_{\tau_D}$, and since $\sup_m \phi'(m) < \infty$ makes ϕ Lipschitz, $\mathbb{E}_{m,z}^\mu[\phi(M_{\tau_d})^+] < \infty$. Hence, τ_d is admissible. ■

Claim A.10. *The stopping time τ_d attains the upper bound:*

$$V(m, z) = \mathbb{E}_{m,z}^\mu[\phi(M_{\tau_d}) - c\tau_d].$$

Proof. Before τ_d , the process remains in the continuation region, and at τ_d , by value matching, $V(M_{\tau_d}, D_{\tau_d}) = \phi(M_{\tau_d})$. The same localization argument as above, now with equality throughout, yields

$$V(m, z) = \mathbb{E}_{m,z}^\mu[V(M_{\tau_d}, D_{\tau_d}) - c\tau_d] = \mathbb{E}_{m,z}^\mu[\phi(M_{\tau_d}) - c\tau_d]. \quad \blacksquare$$

The preceding claims show that τ_d is optimal and that the candidate V coin-

cides with the fixed- ϕ value function. Since the candidate is $C^{1,2}(C_d)$ and satisfies the HJB equation, value matching, smooth fit, and the reflected-boundary condition, this value function is regular for d with terminal payoff ϕ at perceived drift μ .

Finally, suppose $\phi'(m) \rightarrow 1$. Since $d(m) \rightarrow d_\infty$, the right-hand side of the boundary ODE (2) has a finite limit. Since d itself has a finite limit, this limiting derivative must be zero. Hence,

$$0 = 1 - \frac{\mu}{c} \frac{1}{1 - \exp\left(-\frac{2\mu}{\sigma^2} d_\infty\right)},$$

which delivers $d_\infty = \bar{d}_\mu$. ■

As we noted [Proposition A.5](#) shows that any boundary satisfying (2) is optimal. Our next lemma provides the converse within the regular drawdown class: any regular optimal drawdown boundary must satisfy (2). This is the step that will allow the scalar ODE to characterize, rather than merely construct, regular drawdown equilibria.

Given a prize function ϕ , we write

$$V_\phi^\mu(m, z) := \sup_{\tau} \mathbb{E}_{m,z}^\mu [\phi(M_\tau) - c\tau],$$

where the supremum is over stopping times τ satisfying $\mathbb{E}_{m,z}^\mu[\tau] < \infty$ and $\mathbb{E}_{m,z}^\mu[\phi(M_\tau)^+] < \infty$.

Lemma A.11. *Fix $\mu \in (0, c)$. Let $\phi \in C^1([0, \infty))$, and let $d \in C^1([0, \infty))$ be positive. Suppose τ_d is an optimal drawdown rule for the fixed- ϕ problem and that V_ϕ^μ is regular for d with terminal payoff ϕ at perceived drift μ . Then d satisfies*

$$d'(m) = 1 - \frac{\mu}{c} \frac{\phi'(m)}{1 - \exp\left(-\frac{2\mu}{\sigma^2} d(m)\right)}. \quad (3)$$

If, in addition, $\phi'(m) \rightarrow 1$ and $d(m) \rightarrow d_\infty \in (0, \infty)$, then $d_\infty = \bar{d}_\mu$.

Proof of Lemma A.11. Let $V := V_\phi^\mu$ and maintain the shorthand $a := 2\mu/\sigma^2$.

Claim A.12. For each $m \geq 0$ and $0 \leq z \leq d(m)$,

$$V(m, z) = \phi(m) + \frac{c}{\mu} \left[d(m) - z - \frac{1 - \exp(-a(d(m) - z))}{a} \right].$$

Proof. Fix $m \geq 0$. Since V is regular, it satisfies

$$\frac{\sigma^2}{2} V_{zz}(m, z) - \mu V_z(m, z) - c = 0$$

for $0 < z < d(m)$. This is a linear ODE in z . Setting $U(z) := V_z(m, z)$, we get $U'(z) - aU(z) = 2c/\sigma^2$, so $V_z(m, z) = \Gamma(m)e^{az} - c/\mu$ for some function Γ . Moreover, smooth fit, $V_z(m, d(m)^-) = 0$, implies $\Gamma(m) = (c/\mu)e^{-ad(m)}$. Hence,

$$V_z(m, z) = \frac{c}{\mu} [\exp(-a(d(m) - z)) - 1].$$

Integrating from z to $d(m)$ and using value matching, $V(m, d(m)) = \phi(m)$, produces the stated formula for $V(m, z)$. ■

Claim A.13. The boundary d satisfies the free-boundary ODE (3).

Proof. Differentiating the formula for V in the previous claim with respect to m , and then taking $z \downarrow 0$, delivers

$$V_m(m, 0^+) = \phi'(m) + \frac{c}{\mu} d'(m) [1 - \exp(-ad(m))].$$

From the formula for V_z ,

$$V_z(m, 0^+) = -\frac{c}{\mu} [1 - \exp(-ad(m))].$$

Imposing the reflected-boundary condition $V_m(m, 0^+) + V_z(m, 0^+) = 0$, we get

$$0 = \phi'(m) + \frac{c}{\mu} [1 - \exp(-ad(m))] [d'(m) - 1],$$

which we solve for $d'(m)$ to obtain the desired ODE (3). ■

Claim A.14. *If $\phi'(m) \rightarrow 1$ and $d(m) \rightarrow d_\infty \in (0, \infty)$, then $d_\infty = \bar{d}_\mu$.*

Proof. Under these assumptions, the right-hand side of (3) has a finite limit. Since d has a finite limit, this limiting derivative must be zero; otherwise d could not converge. Thus,

$$1 - \exp\left(-\frac{2\mu}{\sigma^2} d_\infty\right) = \frac{\mu}{c},$$

and, therefore, $d_\infty = \bar{d}_\mu$. ■

The three claims prove the lemma. ■

A.3 Proof of Theorem 3.2

Proof of Theorem 3.2. The equation in the theorem is the drawdown equation written along population quantiles. To simplify notation, define its right-hand side by

$$F_\mu(t, x) := x - \frac{\mu}{c} \frac{x + b(t)}{1 - \exp\left(-\frac{2\mu}{\sigma^2} x\right)},$$

so that the equation is $\delta'(t) = F_\mu(t, \delta(t))$. The number $\bar{d}_\mu$ is the limiting drawdown level, characterized by

$$1 - \exp\left(-\frac{2\mu}{\sigma^2} \bar{d}_\mu\right) = \frac{\mu}{c}.$$

Set $B := y'(0)$, $\alpha_\mu := 1 + B/\bar{d}_\mu$, and

$$d_\mu^+ := \frac{\sigma^2}{2\mu} \log\left(\frac{c}{c - \mu\alpha_\mu}\right).$$

Also write $\theta_\mu := \mu/c$, $a_\mu := 2\mu/\sigma^2$, and $g_\mu(x) := 1/(1 - \exp(-a_\mu x))$.

We proceed through six claims. The first four claims solve the scalar terminal-value problem. The last two claims reconstruct the equilibrium and prove uniqueness among regular drawdown equilibria.

Claim A.15. *For every $T > 0$, the finite-terminal problem*

$$\delta'_T(t) = F_\mu(t, \delta_T(t)), \quad \text{with} \quad \delta_T(T) = \bar{d}_\mu \quad (4)$$

has a unique solution on $[0, T]$, and this solution satisfies

$$\bar{d}_\mu \leq \delta_T(t) \leq d_\mu^+ \quad \text{for all } t \in [0, T].$$

Proof. Since $\mu \in \mathcal{M}$, $\theta_\mu \alpha_\mu < 1$, so d_μ^+ is well defined and satisfies $1 - \exp(-a_\mu d_\mu^+) = \theta_\mu \alpha_\mu$. At the lower endpoint,

$$F_\mu(t, \bar{d}_\mu) = -b(t) \leq 0.$$

At the upper endpoint,

$$F_\mu(t, d_\mu^+) = d_\mu^+ - \frac{d_\mu^+ + b(t)}{\alpha_\mu} = \frac{d_\mu^+ (\alpha_\mu - 1) - b(t)}{\alpha_\mu} \geq 0,$$

because $d_\mu^+ \geq \bar{d}_\mu$, $\alpha_\mu - 1 = B/\bar{d}_\mu$, and $b(t) \leq B$.

Since the limiting condition is imposed at infinity, we approximate it with finite terminal problems. Fix $T > 0$, and let us look for a function δ_T on $[0, T]$ satisfying (4). Equivalently, define $h_T(r) := \delta_T(T - r)$. Then $h_T(0) = \bar{d}_\mu$, and h_T solves

$$h'_T(r) = -F_\mu(T - r, h_T(r)).$$

The interval $[\bar{d}_\mu, d_\mu^+]$ cannot be left by this reversed-time equation. At the lower

endpoint, $h_T = \bar{d}_\mu$, we have

$$h'_T(r) = -F_\mu(T - r, \bar{d}_\mu) = b(T - r) \geq 0,$$

so the solution cannot cross below $\bar{d}_\mu$. At the upper endpoint, $h_T = d_\mu^+$, we have

$$h'_T(r) = -F_\mu(T - r, d_\mu^+) \leq 0,$$

so the solution cannot cross above d_μ^+ . Thus, any solution that starts in $[\bar{d}_\mu, d_\mu^+]$ remains in this interval.

The right-hand side of the reversed equation is continuous in r and locally Lipschitz in h_T whenever $h_T > 0$. As a result, the initial-value problem for h_T has a unique local solution. The bounds above keep this solution in the compact interval $[\bar{d}_\mu, d_\mu^+]$, where the right-hand side is bounded. Hence, the solution cannot blow up before time T , so h_T is defined on all of $[0, T]$. Equivalently, $\delta_T(t) := h_T(T - t)$ is uniquely defined on $[0, T]$, solves

$$\delta'_T(t) = F_\mu(t, \delta_T(t)), \quad \text{and} \quad \delta_T(T) = \bar{d}_\mu,$$

and satisfies

$$\bar{d}_\mu \leq \delta_T(t) \leq d_\mu^+ \quad \text{for all } t \in [0, T]. \quad \blacksquare$$

Claim A.16. *There exists a C^1 solution δ_μ of $\delta'_\mu = F_\mu(t, \delta_\mu)$ on $[0, \infty)$, and it satisfies*

$$\bar{d}_\mu \leq \delta_\mu(t) \leq d_\mu^+ \quad \text{for all } t \geq 0.$$

Proof. The family of solutions to (4), $\{\delta_T\}_{T>0}$, is uniformly bounded. It is also equi-Lipschitz on compact time intervals, because F_μ is bounded on $[0, \infty) \times [\bar{d}_\mu, d_\mu^+]$. Choose a sequence $T_n \rightarrow \infty$, and pass to a subsequence, still denoted T_n , such that $T_n \geq n$. Let $n_j^0 := j$.

For each integer $k \geq 1$, discard finitely many terms so that all remaining functions are defined on $[0, k]$, apply the Arzelà-Ascoli theorem on $[0, k]$ to the sequence $(\delta_{T_{n_j^{k-1}}})_j$, and choose a subsequence $(n_j^k)_j$ such that $\delta_{T_{n_j^k}}$ converges uniformly on $[0, k]$. Now set $N_j := n_j^j$. For every finite R , the sequence $\delta_{T_{N_j}}$ converges uniformly on $[0, R]$: if $k_R := \lceil R \rceil$, then for all $j \geq k_R$, N_j belongs to the subsequence chosen for $[0, k_R]$. The limits on overlapping intervals agree, and, therefore, define a continuous function δ_μ on $[0, \infty)$.

For fixed $0 \leq s \leq t$, and all sufficiently large j , the solutions to (4) satisfy

$$\delta_{T_{N_j}}(t) = \delta_{T_{N_j}}(s) + \int_s^t F_\mu(u, \delta_{T_{N_j}}(u)) du.$$

Because $\delta_{T_{N_j}} \rightarrow \delta_\mu$ uniformly on $[s, t]$, and because F_μ is continuous on the compact set $[s, t] \times [\bar{d}_\mu, d_\mu^+]$, the integrands converge uniformly. Passing to the limit along the selected subsequence produces

$$\delta_\mu(t) = \delta_\mu(s) + \int_s^t F_\mu(u, \delta_\mu(u)) du.$$

Thus, δ_μ is C^1 and solves $\delta'_\mu = F_\mu(t, \delta_\mu)$ on $[0, \infty)$. The bounds also pass to the limit, so $\bar{d}_\mu \leq \delta_\mu(t) \leq d_\mu^+$ for all $t \geq 0$. $\blacksquare$

Claim A.17. *The solution δ_μ satisfies $\delta_\mu(t) \rightarrow \bar{d}_\mu$.*

Proof. Since $y'' \leq 0$, $y'(q) \leq y'(0) = B$ for $q \in [0, 1]$, and, hence, $0 \leq b(t) \leq Be^{-t} \leq B$. For $d \in [\bar{d}_\mu, d_\mu^+]$,

$$\partial_d F_\mu(t, d) = 1 - \theta_\mu g_\mu(d) - \theta_\mu (d + b(t)) g'_\mu(d).$$

Since $g'_\mu(d) < 0$, $\theta_\mu g_\mu(d) \leq 1$ for $d \geq \bar{d}_\mu$, and $b(t) \in [0, B]$, continuity on the compact set $[\bar{d}_\mu, d_\mu^+] \times [0, B]$ guarantees $\lambda_\mu > 0$ such that $\partial_d F_\mu(t, d) \geq \lambda_\mu$ for all $t \geq 0$ and all $d \in [\bar{d}_\mu, d_\mu^+]$. Let $x_T(t) := \delta_T(t) - \bar{d}_\mu$. Since $F_\mu(t, \bar{d}_\mu) = -b(t)$, we appeal to the mean-value theorem to deduce $x'_T(t) = a_T(t)x_T(t) - b(t)$ for some $a_T(t) \geq \lambda_\mu$, with

$x_T(T) = 0$. Solving backward,

$$0 \leq x_T(t) \leq \int_t^T \exp\left(-\int_t^s a_T(v)dv\right) b(s)ds \leq \frac{B}{1+\lambda_\mu} e^{-t}.$$

Passing to the limit along the selected subsequence yields $0 \leq \delta_\mu(t) - \bar{d}_\mu \leq B e^{-t}/(1 + \lambda_\mu)$, so $\delta_\mu(t) \rightarrow \bar{d}_\mu$. $\blacksquare$

Claim A.18. *The function δ_μ is the unique bounded positive solution satisfying $\delta(t) \rightarrow \bar{d}_\mu$.*

Proof. First, any bounded positive solution satisfying $\delta(t) \rightarrow \bar{d}_\mu$ must lie in $[\bar{d}_\mu, d_\mu^+]$. If $d < \bar{d}_\mu$, then $1 - \exp(-a_\mu d) < \theta_\mu$, so $F_\mu(t, d) = d(1 - \theta_\mu g_\mu(d)) - \theta_\mu b(t)g_\mu(d) < 0$. Thus, a solution below $\bar{d}_\mu$ moves downward and cannot approach $\bar{d}_\mu$ from below. The same derivative calculation shows F_μ is strictly increasing on $[\bar{d}_\mu, \infty)$. Hence, if $d > d_\mu^+$, then $F_\mu(t, d) > F_\mu(t, d_\mu^+) \geq 0$, so a solution above d_μ^+ cannot move down toward $\bar{d}_\mu$. Therefore, every bounded terminal solution lies in the box.

Now let δ and $\tilde{\delta}$ be two bounded terminal solutions. Their difference $w = \delta - \tilde{\delta}$ satisfies $w'(t) = a(t)w(t)$, where $a(t) > 0$. Hence, for $T > t$,

$$w(t) = w(T) \exp\left(-\int_t^T a(s)ds\right).$$

Since both solutions converge to $\bar{d}_\mu$, $w(T) \rightarrow 0$, and so $w(t) = 0$ for every t . $\blacksquare$

Claim A.19. *The scalar solution δ_μ reconstructs a regular drawdown equilibrium (K_μ, d_μ) .*

Proof. Define $m_\mu(t) := \int_0^t \delta_\mu(s)ds$. Since $\delta_\mu \geq \bar{d}_\mu > 0$, $m_\mu(t) \rightarrow \infty$. Let T_μ be the inverse of m_μ . Define $K_\mu(m) := 1 - e^{-T_\mu(m)}$ and $d_\mu(m) := \delta_\mu(T_\mu(m))$. Then $\int_0^m du/d_\mu(u) = T_\mu(m)$, so [Lemma 3.1](#) implies that K_μ is exactly the true-law distribution induced by d_μ .

Along the quantile path $m_\mu(t)$, $K_\mu(m_\mu(t)) = 1 - e^{-t}$, $K'_\mu(m_\mu(t)) = e^{-t}/\delta_\mu(t)$, and $\phi'_{K_\mu}(m_\mu(t)) = 1 + b(t)/\delta_\mu(t)$. The reconstructed boundary d_μ is C^1 , positive, bounded,

bounded away from zero, and satisfies $d_\mu(m) \rightarrow \bar{d}_\mu$. The reconstructed K_μ is C^1 , so ϕ_{K_μ} is C^1 , increasing, and bounded below. Because $\delta_\mu(t) \geq \bar{d}_\mu$,

$$\frac{\mu}{c} \phi'_{K_\mu}(m_\mu(t)) \leq \frac{\mu}{c} \left(1 + \frac{y'(0)}{\bar{d}_\mu} \right) < 1.$$

The scalar ODE is exactly the fixed- K_μ boundary ODE rewritten along $m_\mu(t)$, since $d'_\mu(m_\mu(t))\delta_\mu(t) = \delta'_\mu(t)$. Hence, d_μ solves (2) with $\phi = \phi_{K_\mu}$. Thus, [Proposition A.5](#) applies: d_μ is optimal given K_μ , and the associated fixed- ϕ_{K_μ} value function is regular for d_μ with terminal payoff ϕ_{K_μ} at perceived drift μ . Since K_μ is generated by d_μ , the pair (K_μ, d_μ) is a regular drawdown equilibrium.

Finally, $K_\mu(m_\mu(t)) = 1 - e^{-t}$, so $m_\mu(t) = K_\mu^{-1}(1 - e^{-t})$, and by construction $m'_\mu(t) = \delta_\mu(t)$. ■

Claim A.20. *Every regular drawdown equilibrium at perceived drift μ is equal to (K_μ, d_μ) .*

Proof. Conversely, let (K, d) be any regular drawdown equilibrium. Set $\phi := \phi_K$. Since d is a regular optimal drawdown boundary for the fixed- K problem at perceived drift μ , τ_d is optimal for the fixed- ϕ problem and the value function $V_K^\mu = V_\phi^\mu$ is regular for d with terminal payoff ϕ at perceived drift μ . By [Lemma A.11](#), d satisfies the fixed- K free-boundary ODE. Since d is bounded above and below, $A_d(m) \rightarrow \infty$, and [Lemma 3.1](#) delivers $K(m) = 1 - \exp(-A_d(m))$. Consequently, K is C^1 , with $K'(m) = \exp(-A_d(m))/d(m) \rightarrow 0$, so $\phi'_K(m) = 1 + y'(K(m))K'(m) \rightarrow 1$. From [Lemma A.11](#), we then have $\lim_{m \rightarrow \infty} d(m) = \bar{d}_\mu$.

Define $m(t)$ as the unique point satisfying $K(m(t)) = 1 - e^{-t}$, and set $\delta(t) = d(m(t))$. Since $K(m) = 1 - \exp(-A_d(m))$, we have $A_d(m(t)) = t$, and so $m'(t) = \delta(t)$. Moreover, $K'(m(t)) = e^{-t}/\delta(t)$, so $\phi'_K(m(t)) = 1 + b(t)/\delta(t)$. Rewriting the fixed- K ODE along $m(t)$ produces the scalar ODE with terminal limit $\bar{d}_\mu$. By the uniqueness of the bounded terminal solution, $\delta = \delta_\mu$. Therefore, $m = m_\mu$, $d = d_\mu$, and $K = K_\mu$. ■

The six claims prove the theorem. ■

B Section 4 Proofs

B.1 Proof of Lemmas 4.1 and 4.2

Proof of Lemma 4.1. First, $\bar{d}_\mu$ is strictly increasing in μ , since

$$\bar{d}_\mu = \frac{\sigma^2 \log\left(\frac{1}{1-\mu/c}\right)}{2c \frac{\mu}{c}},$$

and $x \mapsto \log(1/(1-x))/x$ is strictly increasing on $(0, 1)$.

Let $w(t) := \delta_{\mu_2}(t) - \delta_{\mu_1}(t)$. Since $\bar{d}_{\mu_2} > \bar{d}_{\mu_1}$, $w(t) > 0$ for all sufficiently large t . Suppose, for the sake of contradiction, that $w(t) \leq 0$ for some finite t . Since $w(t) \rightarrow \bar{d}_{\mu_2} - \bar{d}_{\mu_1} > 0$, the set $\{s \geq 0 : w(s) \leq 0\}$ is nonempty and bounded. Let

$$t^* := \sup\{s \geq 0 : w(s) \leq 0\}.$$

By continuity, $w(t^*) = 0$, and $w(t) > 0$ for all $t > t^*$ sufficiently close to t^* .

Let $d := \delta_{\mu_1}(t^*) = \delta_{\mu_2}(t^*)$. Subtracting the two scalar ODEs at t^* , we obtain

$$w'(t^*) = -\frac{d + b(t^*)}{c} \left[\frac{\mu_2}{1 - \exp\left(-\frac{2\mu_2}{\sigma^2}d\right)} - \frac{\mu_1}{1 - \exp\left(-\frac{2\mu_1}{\sigma^2}d\right)} \right].$$

The bracketed term is strictly positive because $x \mapsto x/(1 - e^{-x})$ is strictly increasing on $(0, \infty)$, and so $w'(t^*) < 0$. But at a last crossing before w becomes positive to the right, the right derivative cannot be negative. This contradiction proves $w(t) > 0$ for every finite t . ■

Proof of Lemma 4.2. For every $t \geq 0$,

$$\mathbb{P}(m_\mu(Z) \leq m_\mu(t)) = \mathbb{P}(Z \leq t) = 1 - e^{-t} = K_\mu(m_\mu(t)).$$

Since m_μ is continuous and strictly increasing, the claim follows. $\blacksquare$

B.2 Proof of Theorem 4.3

Proof of Theorem 4.3. By Lemma 4.1, $m'_{\mu_2}(t) > m'_{\mu_1}(t)$ for every finite t . Since $m_{\mu_1}(0) = m_{\mu_2}(0) = 0$, $m_{\mu_2}(t) > m_{\mu_1}(t)$ for every $t > 0$. Lemma 4.2, therefore, implies $M_{\mu_2} \geq_{\text{FOSD}} M_{\mu_1}$, with strict dominance at every positive quantile.

Let $\Delta(t) := m_{\mu_2}(t) - m_{\mu_1}(t)$. Then Δ is increasing and nonconstant. Couple the two stopped maxima using the same $Z \sim \text{Exp}(1)$, and set $X := m_{\mu_1}(Z)$ and $Y := \Delta(Z)$. Then $X \stackrel{d}{=} M_{\mu_1}$ and $X + Y \stackrel{d}{=} M_{\mu_2}$. Since X and Y are increasing functions of the same scalar random variable, $\text{Cov}(X, Y) \geq 0$. Indeed, if Z' is an independent copy of Z , then

$$2\text{Cov}(X, Y) = \mathbb{E}\left[\left(m_{\mu_1}(Z) - m_{\mu_1}(Z')\right)(\Delta(Z) - \Delta(Z'))\right] \geq 0.$$

Since Y is nonconstant, $\text{Var}(Y) > 0$. Also, $\delta_\mu(t) \leq d_\mu^+$, so $m_\mu(t) \leq d_\mu^+ t$, and all second moments are finite. Consequently,

$$\text{Var}^0(M_{\mu_2}) - \text{Var}^0(M_{\mu_1}) = \text{Var}(Y) + 2\text{Cov}(X, Y) > 0. \quad \blacksquare$$

B.3 Proof of Proposition 4.4

Proof of Proposition 4.4. Since d_μ is bounded above and below, $A_\mu(m) \rightarrow \infty$. Hence, $\tau_\mu < \infty$ $\mathbb{P}^0$ -almost surely by Lemma 3.1.

We now compute the joint Laplace transform of τ_μ and M_{τ_μ} . In the notation of Avram, Li, and Li (ALL) (Avram et al., 2021, Theorem 2.1 and Corollary 4.2), the drawdown time is defined by a lower boundary f , with $\tau_f = \inf\{t \geq 0: X_t < f(M_t)\}$. For the present boundary, set $f_\mu(m) := m - d_\mu(m)$, so $m - f_\mu(m) = d_\mu(m)$. Since d_μ satisfies the free-boundary ODE and $\phi'_{K_\mu} > 0$, we have $d'_\mu(m) < 1$, so f_μ is increasing. The weak inequality in the definition of τ_μ and the strict inequality in the

definition of τ_f give the same stopping time almost surely. Indeed, at first contact, $d_\mu(M_t) > 0$, so $X_t < M_t$. Accordingly, M_t is locally constant immediately after contact, and the boundary $f_\mu(M_t)$ is locally constant as well. By the strong Markov property and the nonstickiness of Brownian motion at a point, the process crosses the boundary immediately.

Specializing the ALL formula to driftless Brownian motion with volatility σ , using $\lambda(q) := \sqrt{2q}/\sigma$, $\varphi_q^+(x) = e^{\lambda(q)x}$ and $\varphi_q^-(x) = e^{-\lambda(q)x}$, produces

$$\mathbb{E}^0 \left[e^{-q\tau_\mu}; M_{\tau_\mu} \in dm \right] = \frac{\lambda(q)}{\sinh(\lambda(q)d_\mu(m))} \exp \left[- \int_0^m \lambda(q) \coth(\lambda(q)d_\mu(u)) du \right] dm.$$

As a check, when $d_\mu(m) \equiv D$, integrating this density against $e^{-\beta m}$ gives us

$$\mathbb{E}^0 \left[e^{-q\tau_D - \beta M_{\tau_D}} \right] = \frac{\lambda(q)}{\lambda(q) \cosh(\lambda(q)D) + \beta \sinh(\lambda(q)D)},$$

which is the classical fixed-drawdown Brownian formula of Taylor and Williams, and the fixed-drawdown specialization of Lehoczky's maximum-based formula [Taylor \(1975\)](#); [Williams \(1976\)](#); [Lehoczky \(1977\)](#).

Letting $q \downarrow 0$ in the joint Laplace density yields

$$\mathbb{P}^0(M_{\tau_\mu} \geq m) = e^{-A_\mu(m)}, \quad \text{and} \quad \mathbb{P}^0(M_{\tau_\mu} \in dm) = \frac{e^{-A_\mu(m)}}{d_\mu(m)} dm.$$

The mean stopping time follows by differentiating the Laplace transform at zero. With $\lambda = \sqrt{2q}/\sigma$,

$$\lambda \coth(\lambda d) = \frac{1}{d} + \frac{\lambda^2 d}{3} + O(\lambda^4 d^3), \quad \text{and} \quad \frac{\lambda}{\sinh(\lambda d)} = \frac{1}{d} \left(1 - \frac{\lambda^2 d^2}{6} + O(\lambda^4 d^4) \right).$$

Since d_μ is bounded above and below, these expansions are uniform for d in the range of d_μ . For q near zero, the resulting densities and their first q -derivatives are dominated by $C(1+m)\exp(-m/C)$ for some finite C . Therefore, we can differenti-

ate the expansion at $q = 0$ and integrate over m . Substituting, we have

$$\mathbb{E}^0[\tau_\mu; M_{\tau_\mu} \in dm] = \frac{e^{-A_\mu(m)}}{3\sigma^2 d_\mu(m)} \left[d_\mu(m)^2 + 2 \int_0^m d_\mu(u) du \right] dm.$$

Then divide by the density of M_{τ_μ} to get

$$\mathbb{E}^0[\tau_\mu | M_{\tau_\mu} = m] = \frac{1}{3\sigma^2} \left[d_\mu(m)^2 + 2 \int_0^m d_\mu(u) du \right],$$

where the conditional expectation is understood as the Radon-Nikodym derivative of $\mathbb{E}^0[\tau_\mu; M_{\tau_\mu} \in dm]$ with respect to the law of M_{τ_μ} .

Now let $B_\mu(m) := \int_0^m d_\mu(u) du$ and $S_\mu(m) := e^{-A_\mu(m)}$. Since $S'_\mu(m) = -S_\mu(m)/d_\mu(m)$, we integrate by parts to obtain

$$\int_0^\infty B_\mu(m) \frac{S_\mu(m)}{d_\mu(m)} dm = \int_0^\infty d_\mu(m) S_\mu(m) dm,$$

where the boundary term vanishes because $B_\mu(m)$ grows at most linearly and $S_\mu(m)$ decays exponentially. In sum,

$$\mathbb{E}^0[\tau_\mu] = \frac{1}{\sigma^2} \int_0^\infty d_\mu(m) e^{-A_\mu(m)} dm = \frac{1}{\sigma^2} \mathbb{E}^0[d_\mu(M_{\tau_\mu})^2] = \frac{1}{\sigma^2} \int_0^\infty \delta_\mu(t)^2 e^{-t} dt,$$

where the second equality uses the density of M_{τ_μ} and the third equality follows from the fact that in hazard time, $A_\mu(m_\mu(t)) = t$, $dm_\mu(t) = \delta_\mu(t) dt$, and $d_\mu(m_\mu(t)) = \delta_\mu(t)$.

We conclude that if $\mu_1 < \mu_2$, then $\delta_{\mu_2}(t) > \delta_{\mu_1}(t)$ for every finite t by [Lemma 4.1](#) and so $\mathbb{E}^0[\tau_{\mu_2}] > \mathbb{E}^0[\tau_{\mu_1}]$. ■